CYBERCRIME CASE STUDY · 2025

RANSOM WARE

Digital extortion. Encryption-based attacks.
The real human cost of cybercrime.

An in-depth analysis covering attack mechanics, vulnerable targets, documented consequences, and the WannaCry incident of 2017.

CONTENTS

- What Is Ransomware?
- How an Attack Unfolds ?
- Who Gets Hit ?
- Consequences Real Incident — WannaCry 2017

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01

WHAT IS RANSOMWARE?

Put simply, ransomware locks you out of your own files. It gets onto a device, encrypts everything — money before giving access back. The encryption is strong enough that there is no practical way around it

Payment is almost always in cryptocurrency — Bitcoin mostly, without the attacker's key because it is harder to trace. Victims get a tight deadline, usually 48 to 72 hours, after which the price increases or the files are deleted. Some attackers also threaten to publish stolen data publicly if payment is not made.

What makes ransomware especially hard to fight is that it has become a service. Groups called Ransomware-as-a-Service operators build the malware and rent it out to other criminals, who handle the actual attacks and share the profits. You no longer need technical ability to run an attack — just money to buy access.

Unlike most malware that stays hidden, ransomware is deliberately visible. It plasters a ransom note across every screen on the network. The countdown timer, the threats, the helplessness — all of it is calculated to cause panic and force a fast decision.

> "A criminal with no coding skills can buy a ransomware kit online and have a target infected by morning."

RaaS Model

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SECTION 02

How an Attack Unfolds

Six systematic phases — from the first click to the ransom note.

01

Getting In

Phishing emails remain the most common entry — an invoice, an HR notice, a delivery alert that looks real. Other routes: unpatched software flaws like EternalBlue, or exposed remote desktop services with weak passwords.

02

Setting Up Quietly

The malwarerunsin the background, disabling security tools, making itself persistent across reboots, and calling back to the attacker's server to collect the encryption keys it will need later.

03

Spreading Across the Network

Thisis where one infected machine becomes an organisation-wide crisis. The malware scans for reachable devices, jumps across shared drives and servers, and keeps moving until it has covered as much ground as possible.

04

Stealing Data First

Newergroupscopysensitive files to their own servers before triggering encryption. This gives them a second card: pay the ransom, or watch your data appear on public leak sites.

05

Encryption

Filesare locked using AES-256 — the same algorithm banks use. Extensions get changed to .WNCRY or .locked. On a large network this can run silently for hours.

06

The Ransom Note

Every screen displaysthe same message: a Bitcoin wallet address, a timer counting down, and a threat. Everything is engineered to create maximum pressure and push a panicked, quick decision.

SECTION 03

Who Gets Hit?

Attackers go wheredesperationishighest.Organisations that cannot survive downtime are far morelikely to pay.

- Ø<ßå Hospitals
- Ø<ßÛþGovernment.
- Øt<ßë Universities
- Ø<ßí Manufacturing
- &¡Infrastructure
- Ø<ßâ Mid-size Businesses
- Ø=Ü° Finance
- Ø=Üd Home Users

SECTION 04 — CONSEQUENCES

What Is the Fallout?

EverythingStops
Work halts completely — no files, no systems. Sometimes for weeks.

CostsSpiral
Recovery expenses often dwarf the ransom itself. Millions, routinely.

Data Leaks Anyway
Double extortion means stolen records appear online regardless of payment.

Reputation Damage
A public ransomware incident takes years to fully recover from.

Fines Follow
GDPR and similar laws add heavy penalties on top of recovery costs.

Lives at Risk
Hospital outages have delayed surgeries. The human cost is documented.

WANNACRY

RansomwareAttacK

Global Incident · May 2017

On 12 May 2017, malware called WannaCry spread across the internet faster than anything before it. Within hours it had reached over 150 countries. By end of day — more than 200,000 machines locked and unusable.

The key was a Windows vulnerability called EternalBlue, discovered by the US NSA, kept secret for years, then leaked online by a group called the Shadow Brokers. WannaCry needed no user action — it scanned networks automatically and jumped device to device on its own.

The UK's NHS suffered badly. Hospitals lost patient records. Thousands of appointments were cancelled, surgeries postponed, ambulances redirected. A British researcher found a dormant kill-switch domain; registering it for under £10 halted the spread — but recovering encrypted data was almost impossible for those already hit.

The patch that would have prevented everything — MS17-010 — had been available for two full months. Most affected systems had never applied it. That single gap caused billions in damage and put real lives in danger.

BY THE NUMBERS

200K+

150

Countries Hit

$4B+

Global Damage

£10

Kill-Switch Cost

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